Generated by Scopus AI, Sat Oct 10 2026

Fishermen crossing the Imbl and consequences

Preventing Fishermen from Crossing the International Maritime Boundary Line (IMBL): Current Situation, Consequences, Enforcement, and Effective Strategies

Quick Reference

Key Findings Table

Theme	Current Situation	Existing Measures	Gaps/Limitations	More Effective Steps
Patterns & Drivers	Frequent, often intentional crossings due to economic, environmental, and socio-cultural pressures; especially acute in South Asia (Palk Bay)	Limited navigation aids, some GPS/NAVIC alert systems	Socio-economic vulnerability, poor awareness, weak enforcement	Integrated tech (ML, Li-Fi), community empowerment, regional cooperation
Consequences	Arrest, detention, boat seizure, poverty, food insecurity, human rights violations	Legal penalties, punitive enforcement, some support programs	Human rights concerns, lack of legal harmonization, social marginalization	Rights-based enforcement, social safety nets, legal reforms
Enforcement	Fragmented, inconsistent, often reactive	GPS/NAVIC alerts, MCS frameworks, regional agreements	Poor inter-agency coordination, legal gaps, limited tech coverage	Harmonized frameworks, data sharing, best practices from RFMOs
Prevention	Tech-based alerts, some community programs	GPS, IoT, GSM, limited ML pilots	Cost, coverage, user adoption, lack of integration	ML-based predictive systems, Li-Fi, economic alternatives, participatory governance

Direct Answer

Current measures to prevent accidental and intentional fishing across the IMBL include GPS-based and NAVIC-driven alert systems, real-time vessel tracking via IoT and GSM networks, and integrated MCS frameworks designed to warn fishermen when approaching maritime boundaries. However, enforcement remains inconsistent due to fragmented inter-agency coordination and varying legal frameworks. More effective steps involve deploying advanced machine learning models such as DODGE-SAFE for predictive analytics, integrating Li-Fi with GPS

systems for secure, high-speed communication, enhancing community-based economic programs to reduce dependency on risky fishing practices, and fostering regional cooperation to harmonize legal standards and enforcement protocols.

Study Scope

- **Time Period:** 2010–2023, with emphasis on recent (2015–2023) developments
- **Disciplines:** Fisheries management, maritime law, socio-economics, technology (IoT, ML, Li-Fi), human rights, policy studies
- **Methods:** Meta-analysis of empirical studies, policy reviews, case studies, and technical evaluations

Assumptions & Limitations

- Most data are from South Asia (notably India, Sri Lanka, Bangladesh), but findings may generalize to other developing maritime regions.
- Limited availability of detailed legal case studies on individual arrests.
- Technological effectiveness is often based on pilot studies or simulations, not always on large-scale deployments.
- Socio-cultural insights are context-dependent and may not capture all local nuances.

Suggested Further Research

- Cost-effectiveness and user adoption studies for integrated multi-technology systems.
- Detailed legal and human rights case documentation.
- Longitudinal impact assessments of community-based economic empowerment programs.
- Comparative studies of enforcement models across regions.

1. Introduction

Background and Scope

Fishermen crossing the International Maritime Boundary Line (IMBL) is a persistent and complex issue, particularly in regions like the Palk Bay between India and Sri Lanka. These crossings—often driven by economic necessity, environmental degradation, and traditional fishing rights—expose fishermen to severe legal, economic, and human rights consequences. Despite technological and policy interventions, accidental and intentional crossings continue, underscoring the need for more effective, integrated prevention strategies that balance enforcement with the socio-economic realities of coastal communities [1] [2] [3].

2. Current Situation of Fishermen Crossing the IMBL

Patterns and Motivations

- **Frequency & Nature:** Crossings are frequent, with a mix of accidental (due to poor visibility, lack of navigation aids) and intentional (economic necessity, richer fishing grounds) incidents. In the Palk Bay, most crossings are intentional, driven by resource access and economic pressures [1] [2] [4] [5] [6].
- **Drivers:**
  - **Economic:** Declining fish stocks, overfishing, and limited local fishing grounds push fishermen to risk crossing boundaries [4] [7].
  - **Environmental:** Degradation (e.g., coral reef loss), climate variability, and increased competition exacerbate resource scarcity [8] [9] [10].
  - **Socio-cultural:** Traditional rights, kinship networks, and community norms often override legal boundaries [3] [11] [12].

Environmental and Economic Drivers

- **Environmental Degradation:** Overfishing, coral reef depletion, and climate change reduce local fish stocks, forcing fishermen to seek alternative grounds, sometimes across the IMBL [8] [9] [10].
- **Fish Aggregating Devices (FADs):** Their use increases fishing efficiency but can lead to unsustainable practices and boundary violations. Regulation is needed to limit ecological and economic harm [13] [14] [15].
- **Economic Pressures:** Poverty, lack of alternative livelihoods, and indebtedness are strong motivators for risky crossings [16] [17].

Socio-Cultural Perceptions and Community Dynamics

- **Risk Acceptance:** Many fishermen accept high risks as necessary for family survival, influenced by social standing and occupational norms [18] [19].
- **Traditional Rights:** Disputes over traditional fishing grounds are common, with some local agreements attempted but often overridden by national interests [11] [12].
- **Community Involvement:** There is a strong desire for participatory governance and local input in policy-making [20] [21].

Synthesis: The drivers of IMBL crossings are deeply intertwined—economic, environmental, and socio-cultural factors reinforce each other, making simple enforcement insufficient without addressing root causes [4] [20] [22].

3. Consequences Faced by Fishermen

Legal and Economic Consequences

- **Legal Penalties:** Arrest, detention, boat seizure, and sometimes violence (including shooting or sinking of vessels) are common punitive measures [3] [23] [24] [25] [26].
- **Economic Losses:** Loss of boats and gear, fines, and inability to fish during detention periods lead to severe economic hardship [25] [26].
- **IUU Fishing:** Illegal, unreported, and unregulated fishing exacerbates resource depletion and economic insecurity [24] [25].

Social and Human Rights Impacts

- **Poverty and Food Insecurity:** Families of detained fishermen face acute poverty and food insecurity, with women and children disproportionately affected [27] [28] [29].
- **Human Rights Violations:** Detention without trial, violence, and lack of due process are documented, raising international human rights concerns [2] [29] [30].
- **Social Marginalization:** Arrests and detentions disrupt family structures, erode social capital, and can force occupational shifts [2] [12] [31] [32].

Socio-Political and Community Effects

- **Ethnic and Geopolitical Tensions:** Enforcement actions can exacerbate ethnic tensions and complicate diplomatic relations [12] [32].
- **Community Vulnerability:** Marginalized communities face compounded risks due to weak governance and lack of social safety nets [33] [34].

Synthesis: The consequences of IMBL crossings are severe and multidimensional, affecting not just individual fishermen but entire communities and regional stability [2] [27] [30].

4. Enforcement and Policy Measures

Technological Enforcement Mechanisms

- **GPS/NAVIC Alert Systems:** Real-time tracking and alerting systems warn fishermen as they approach boundaries; some systems include automatic vessel control (e.g., speed reduction) and emergency features [1] [2] [35] [36].
- **IoT, GSM, LoRa, Li-Fi:** These technologies enhance communication and alert delivery, especially in areas with poor cellular coverage [37] [38].
- **Coverage Limitations:** Signal availability, high costs, and infrastructure gaps limit effectiveness, especially for small-scale fishers [2] [37].

Policy and Governance Frameworks

- **MCS Frameworks:** Monitoring, Control, and Surveillance systems exist but are hampered by fragmented governance, poor inter-agency coordination, and inconsistent implementation [39] [40].
- **Legal Gaps:** National laws often lack specificity or harmonization, and enforcement is complicated by overlapping jurisdictions [41] [42].
- **Data Sharing:** Limited integration and information sharing among agencies reduce enforcement effectiveness [40] [43].

Regional Cooperation and Best Practices

- **Regional Agreements:** Some success in joint patrols, port state measures, and information exchange, but legal incoherence and limited capacity persist [44] [45] [46].
- **RFMOs:** Regional Fisheries Management Organizations provide models for compliance, periodic performance reviews, and harmonized enforcement, though challenges remain in inclusivity and decision-making [47] [48] [49].

Synthesis: While technological and policy frameworks exist, their effectiveness is undermined by fragmented implementation, legal gaps, and limited regional cooperation. Best practices from RFMOs and integrated tech solutions offer pathways for improvement [39] [47].

5. Prevention Strategies and Research Gaps

Technological Prevention Strategies

- **Machine Learning (ML) Systems:** Advanced models like DODGE-SAFE provide predictive, context-aware guidance, outperforming traditional alert systems in accuracy and prevention rates [50] [51].
- **Li-Fi Integration:** Offers high-speed, secure, and interference-free communication, especially effective for localized, real-time alerts when combined with GPS [37] [51] [52].
- **IoT and Sensor Fusion:** Edge-intelligent buoys and multi-modal sensors enable decentralized, real-time monitoring and alerting [2] [53] [54].
- **Automatic Vessel Control:** Systems that physically prevent boundary crossing (e.g., engine speed reduction) add a layer of enforcement [35] [55] [56] [57] [58].

Community-Based and Economic Empowerment Approaches

- **Alternative Livelihoods:** Programs in the Philippines and Indonesia (e.g., aquaculture, processing, cooperative strengthening) have reduced illegal crossings and improved incomes [20] [22] [59].
- **Participatory Governance:** Involving communities in policy-making and monitoring increases compliance and addresses root causes [60] [61].

- **Women's Empowerment:** Targeted interventions improve household resilience and reduce dependency on risky fishing [62] [63].

Integrated Multi-Technology Systems and Implementation Barriers

- **Barriers:** Institutional fragmentation, infrastructure gaps, interoperability issues, and lack of training hinder adoption [35] [64] [65] [66] [67].
- **Technical Challenges:** Integrating diverse sensors, ensuring reliable communication, and retrofitting existing vessels are significant hurdles [35] [68] [69].
- **Institutional Challenges:** Limited stakeholder involvement, resource constraints, and political conflicts impede implementation [66] [67] [70].

Research Gaps and Future Directions

- **Cost-Effectiveness:** Need for scalable, affordable solutions tailored to small-scale fisheries [37] [71].
- **Legal Harmonization:** More research on harmonizing national and international legal frameworks to protect fishermen's rights [29] [41] [72].
- **User Adoption:** Studies on behavioral and cultural factors influencing technology uptake are lacking [73] [74].

Synthesis: Prevention requires a multi-pronged approach—advanced technology, community engagement, and legal reform. Research must address integration, affordability, and user-centered design [22] [50] [64].

6. Recommendations for More Effective Prevention

Technological Innovations and Deployment

- **Deploy ML-Based Predictive Systems:** Scale up models like DODGE-SAFE for real-time, context-aware boundary alerts and route optimization [50] [51].
- **Integrate Li-Fi with GPS:** Combine high-speed, secure Li-Fi communication with GPS/NAVIC for robust, localized alerting, especially in areas with poor RF coverage [37] [52].
- **Automatic Vessel Control:** Implement systems that can physically prevent boundary crossing when warnings are ignored [35] [55] [56] [57] [58].
- **Emergency Features:** Ensure all systems include SOS and environmental monitoring for comprehensive safety [2] [38].

Policy and Governance Enhancements

- **Harmonize Legal Frameworks:** Align national laws with international standards, clarify penalties, and ensure due process for detained fishermen [39] [47].

- **Improve Inter-Agency Coordination:** Establish formal data sharing and joint enforcement protocols among maritime agencies [40] [43].
- **Adopt Best Practices from RFMOs:** Regular performance reviews, stakeholder representation, and transparent compliance mechanisms [47] [49].

Community Engagement and Economic Support

- **Participatory Governance:** Involve fishing communities in policy design, monitoring, and enforcement [20] [22] [60].
- **Economic Empowerment:** Expand alternative livelihood programs, cooperative models, and women's empowerment initiatives [59] [61].
- **Social Safety Nets:** Provide support for families affected by detention or economic loss due to enforcement actions [2] [27].

Synthesis: Effective prevention is only possible through integrated technological, policy, and community-based interventions, tailored to local contexts and supported by robust legal and institutional frameworks [39] [50] [60].

7. Conclusion

Summary and Outlook

The challenge of preventing fishermen from crossing the IMBL is multifaceted, rooted in economic necessity, environmental change, and socio-cultural dynamics. While current measures—GPS/NAVIC alerts, MCS frameworks, and regional agreements—offer partial solutions, their effectiveness is limited by fragmented enforcement, legal gaps, and socio-economic vulnerability. Advanced technological systems (ML, Li-Fi), harmonized policy frameworks, and community-based economic empowerment are essential for sustainable, humane, and effective prevention. Future research should prioritize cost-effective, user-friendly integrated systems, legal harmonization, and participatory governance to ensure the safety and resilience of fishing communities [1] [3].

References

1. Lifeline system for fisherman Sai Charan, A., Srinivasaverma, V.S.M., Noor Mahammad, S. Lecture Notes in Electrical Engineering, 2016 https://www.scopus.com/pages/publications/84954185486?origin=scopusAI

2. Fishing Vessel Alert System Using NAVIC Vallikannu, R., Sai Revanth, P., Abhishek Reddy, G., Srinivasan, R. 2023 World Conference on Communication and Computing, WCONF 2023, 2023 https://www.scopus.com/pages/publications/85173022293?origin=scopusAI

3. Automatic border alert system for fishermen using GPS and GSM techniques Ranjith, S., Shreyas, Pradeep Kumar, K., Karthik, R. Indonesian Journal of Electrical Engineering and Computer Science, 2017 https://www.scopus.com/pages/publications/85027589244?origin=scopusAI

4. Palk Bay fishing problem requires Indo-Sri Lankan joint-governance Vincent, S. Maritime Affairs, 2020 https://www.scopus.com/pages/publications/85096122545?origin=scopusAI

5. A place to live and fish: Relational place making among the trawl fishers of Palk Bay, India Stephen, J. Ocean and Coastal Management, 2014 https://www.scopus.com/pages/publications/84908135699?origin=scopusAI

6. The Political Ecology of Palk Bay Fisheries: Geographies of Capital, Fisher Conflict, Ethnicity and Nation-State Menon, A., Bavinck, M., Stephen, J., Manimohan, R. Antipode, 2016 https://www.scopus.com/pages/publications/84957431812?origin=scopusAI

7. Maritime piracy in southeast Asia: Consequences on economic and political integration Morabito, G., Di Fatta, D., D'Aleo, V., (...), Vesperi, W. Establishing Food Security and Alternatives to International Trade in Emerging Economies, 2017 https://www.scopus.com/pages/publications/85028885726?origin=scopusAI

8. The Influence of Coastal and Marine Ecosystem Conditions on Fisheries and Socio-Economic Activities of Local Fishermen Hidayati, D. IOP Conference Series: Earth and Environmental Science, 2021 https://www.scopus.com/pages/publications/85104870219?origin=scopusAI

9. Climate-Driven Dynamics and Adaptive Governance Production of Caesio cuning in the Coral Triangle: A Case Study from Karimunjawa, Indonesia Putra, P.C.P., Wahyudi, S.T., Sambah, A.B., Sartimbul, A. Egyptian Journal of Aquatic Biology and Fisheries, 2026 https://www.scopus.com/pages/publications/105051581072?origin=scopusAI

10. Impacts of climate change and ocean acidification on coral reef fisheries: An integrated ecological-economic model Speers, A.E., Besedin, E.Y., Palardy, J.E., Moore, C. Ecological Economics, 2016 https://www.scopus.com/pages/publications/84967194628?origin=scopusAI

11. Impact and Recognition of Traditional Fishing Rights in Maritime Boundary Delimitation: Overview of State Practice in Latin America and Africa Enyew, E.L. Ocean Development and International Law, 2025 https://www.scopus.com/pages/publications/105015846655?origin=scopusAI

12. Transboundary Dialogues and the 'Politics of Scale' in Palk Bay Fisheries: Brothers at Sea? Stephen, J., Menon, A., Scholtens, J., Bavinck, M. South Asia Research, 2013 https://www.scopus.com/pages/publications/84878102509?origin=scopusAI

13. Data Provision for Science-Based FAD Fishery Management: Spanish FAD Management Plan as a Case Study Báez, J.C., Déniz, S., Ramos, M.L., (...), Abascal, F. Sustainability (Switzerland), 2022 https://www.scopus.com/pages/publications/85126305815?origin=scopusAI

14. Utilization of Marine Geospatial Data for Determining Optimal FAD Locations in Improving the Living Standards of the North Gorontalo Coastal Community Djunarsjah, E., Julian, M.M., Muhammad, F., (...), Pamungkas, B. Sustainability (Switzerland), 2023 https://www.scopus.com/pages/publications/85148016710?origin=scopusAI

15. Fish aggregating devices in the eastern tropical Pacific marine corridor, according to the Colombian Fisheries Observer Program Isaza-Toro, E., Selvaraj, J.J., Giraldo-Lopez, A., Ortíz-Ferrín, O.O.Y. Aquatic Conservation: Marine and Freshwater Ecosystems, 2021 https://www.scopus.com/pages/publications/85112048140?origin=scopusAI

16. A case study of illegal fishing causes during seasonal fishery closure in Kaptai Lake, Bangladesh Shalehin, M.S., Parvez, M.T., Lucas, M.C., Galib, S.M. Fisheries Management and Ecology, 2022 https://www.scopus.com/pages/publications/85125408993?origin=scopusAI

17. Socio economic status of fishermen community, south Punjab, Pakistan Hanif, M.S., Iqbal, K.J., Javid, A., (...), Altaf, M. Punjab University Journal of Zoology, 2019 https://www.scopus.com/pages/publications/85077866862?origin=scopusAI

18. Survival at a cost: how artisanal fishers perceive occupational hazards in the Okavango Delta, Botswana Kolawole, O.D., Bolobilwe, K. South African Geographical Journal, 2019

https://www.scopus.com/pages/publications/85055647569?origin=scopusAI

19. The link between risk taking, fish catches, and social standing: Untangling a complex cultural landscape Naranjo-Madrigal, H., van Putten, I. Marine Policy, 2019 https://www.scopus.com/pages/publications/85057553090?origin=scopusAI

20. Involving local fishing communities in policy making: Addressing Illegal fishing in Indonesia Chapsos, I., Koning, J., Noortmann, M. Marine Policy, 2019 https://www.scopus.com/pages/publications/85073037319?origin=scopusAI

21. Community-based and co-management institutions for sustainable coastal fisheries management in Southeast Asia Pomeroy, R.S. Ocean and Coastal Management, 1995 https://www.scopus.com/pages/publications/0029541202?origin=scopusAI

22. SUSTAINABLE LIVELIHOODS AND COASTAL RESILIENCE: LESSONS FROM FISHERFOLK COMMUNITIES IN DUMARAN AND ARACELI Plasus, L.N., Pentang, J.T., Rosero, M.C.C., (...), Plasus, M.M.G. Journal of Integrated Coastal Zone Management, 2025 https://www.scopus.com/pages/publications/105031072145?origin=scopusAI

23. Arduino based Maritime Border Alert System for Fisherman Jacob Varghese, L., Sakthivel, A., Saran Kirthic, J.K., (...), Shri Pavithra, R. 5th International Conference on Inventive Computation Technologies, ICICT 2022 - Proceedings, 2022 https://www.scopus.com/pages/publications/85137316948?origin=scopusAI

24. International legal arrangements for illegal, unreported, and unregulated fishing: legal mechanisms and challenges in Bangladesh Islam, M.S., Hasan, A.S.M.M. Marine Development, 2024 https://www.scopus.com/pages/publications/105022285624?origin=scopusAI

25. Governance of illegal, unreported, and unregulated (IUU) fishing in Bangladesh: status, challenges, and potentials Mozumder, M.M.H., Uddin, M.M., Schneider, P., (...), Nur, A.-A.U. Frontiers in Marine Science, 2023 https://www.scopus.com/pages/publications/85164494287?origin=scopusAI

26. Social-ecological vulnerability to climate change and risk governance in coastal fishing communities of Bangladesh Sultana, R., Irfanullah, H.M., Selim, S.A., Alam, M.S. Frontiers in Marine Science, 2023 https://www.scopus.com/pages/publications/85178966477?origin=scopusAI

27. Assessment of food insecurity and food coping strategies among fishermen household during monsoon in Terengganu, Malaysia Sanusi, N.N.N.N.M., Ali, A., Kamarudin, K.S., Yusof, H.M. Malaysian Applied Biology, 2018 https://www.scopus.com/pages/publications/85060351199?origin=scopusAI

28. Determinants and Consequences of Food Insecurity in Artisanal Fishing Families From the Coastal Community of Sonora, Mexico Quizán-Plata, T., Carrasco-Miranda, J.G., Murillo-Castillo, K.D., (...), Frongillo, E.A. Food and Nutrition Bulletin, 2020 https://www.scopus.com/pages/publications/85092898426?origin=scopusAI

29. The implementation of vessel-sinking policy as an effort to protect indonesian fishery resources and territorial waters Nurdin, Ikaningtyas, Kurniaty, R. IOP Conference Series: Earth and Environmental Science, 2018 https://www.scopus.com/pages/publications/85046109702?origin=scopusAI

30. Fishing for justice: Human rights, development, and fisheries sector reform Ratner, B.D., Åsgård, B., Allison, E.H. Global Environmental Change, 2014 https://www.scopus.com/pages/publications/84905184804?origin=scopusAI

31. Fishing in the troubled water: Media framing of the human rights violations at Palk Bay Ravi Krishnan, C.J.R., Pichaandy, C. Media Watch, 2018 https://www.scopus.com/pages/publications/85044275438?origin=scopusAI

32. Small States and the Politics of Maritime Boundaries in South Asia: Case Studies of India–Sri Lanka and India–Bangladesh Maritime Boundaries Das, U. Geopolitics, 2026

https://www.scopus.com/pages/publications/105032120862?origin=scopusAI

33. Promoting fishing profession and general well-being: A call for labour policy Rashid, M.M., Jamir Singh, P.S., Azman, A. Cogent Social Sciences, 2023 https://www.scopus.com/pages/publications/85150950523?origin=scopusAI

34. Socio-economic Status of Fishermen: A Case Study in the Bangali River Region, Bangladesh Hasan, M.S., Haque, S.A., Enam, S., (...), Al Jufaili, S.M. International Journal of Agriculture and Biosciences, 2025 https://www.scopus.com/pages/publications/105017649919?origin=scopusAI

35. Integrated Sea-Way Alert and Weather Monitoring System Vijayalakshmi, V.J., Vanitha, U., Karshimaa, G., (...), Yaswanth, K.M. 3rd International Conference on Emerging Research in Computational Science, ICERCS 2025, 2025 https://www.scopus.com/pages/publications/105044553611?origin=scopusAI

36. An Improved GPS Assisted Fishermen Tracking System Using Internet of Things with Smart Sensors Association Karthik, G.M., Samuthira Pandi, V., Gayathri, S., (...), Sujatha, S. 2024 4th Asian Conference on Innovation in Technology, ASIANCON 2024, 2024 https://www.scopus.com/pages/publications/85218125701?origin=scopusAI

37. LiFi: A Visible Light Communication Assisted Fishermen Tracking System using GPS Chandran, K.P., Niji, P.S., Chinnammal, V., (...), Jayashree, N. Proceedings of the 2nd IEEE International Conference on Advances in Computing, Communication and Applied Informatics, ACCAI 2023, 2023 https://www.scopus.com/pages/publications/85170030483?origin=scopusAI

38. LoRa-Enabled Intelligent Maritime Safety and Real-Time Fisherman Tracking System using IoT-based Alert Communication Framework Shanmugavadivel, G., Karthi, As., Khalandhar, A., (...), Mouli, S. Proceedings of 4th International Conference on Electronics and Renewable Systems, ICEARS 2026, 2026 https://www.scopus.com/pages/publications/105035737093?origin=scopusAI

39. Bridging policy and practice: A review of monitoring, control, and surveillance in managing illegal fishing in India Dudi, A., Collins, C., Nuno, A., Karnad, D. Marine Policy, 2025 https://www.scopus.com/pages/publications/105014251711?origin=scopusAI

40. The state of maritime law enforcement of four Southeast Asian disputants in the South China Sea Nguyen, H.-K.T., Sato, Y., Ho, M.-T. WMU Journal of Maritime Affairs, 2026 https://www.scopus.com/pages/publications/105051397526?origin=scopusAI

41. The law of the sea Churchill, R., Lowe, V., Sander, A. The law of the sea, 2022 https://www.scopus.com/pages/publications/85168480561?origin=scopusAI

42. Evaluating the potential for transboundary management of marine biodiversity in the Western Indian Ocean Levin, N., Beger, M., Maina, J., (...), Kark, S. Australasian Journal of Environmental Management, 2018 https://www.scopus.com/pages/publications/85041489825?origin=scopusAI

43. Maritime information sharing environment deployment using the advanced multilayered Data Lake capabilities: EFFECTOR project case study Paladin, Z., Kapidani, N., Scrima, P., (...), Astyakopoulos, A. Pomorstvo, 2022 https://www.scopus.com/pages/publications/85144485672?origin=scopusAI

44. Regional patterns in coordination and cooperation in the fight against illegal, unreported, and unregulated fishing Auld, K., Okafor-Yarwood, I., Geymonat, S.J., (...), Neat, F. WMU Journal of Maritime Affairs, 2025 https://www.scopus.com/pages/publications/105015057081?origin=scopusAI

45. Enhancing responsible fishing practices in South East Asia to combat Illegal, Unreported and Unregulated (IUU) fishing Johns, M. Australian Journal of Maritime and Ocean Affairs, 2013 https://www.scopus.com/pages/publications/85087429762?origin=scopusAI

46. To our children's children's children: From promoting to achieving compliance in high seas fisheries Rayfuse, R. International Journal of Marine and Coastal Law, 2005
https://www.scopus.com/pages/publications/33746624748?origin=scopusAI

47. The impact of Regional Fishery Management Organization structure on collaborative performance Roozee, E., Temby, O., Hickey, G.M. Ocean and Coastal Management, 2026
https://www.scopus.com/pages/publications/105043065767?origin=scopusAI

48. The influence of performance reviews on regional fisheries management organizations Haas, B., Haward, M., McGee, J., (...), Hoel, A.H. ICES Journal of Marine Science, 2019
https://www.scopus.com/pages/publications/85083800341?origin=scopusAI

49. Performance of regional fisheries management organizations: Ecosystem-based governance of bycatch and discards Gilman, E., Passfield, K., Nakamura, K. Fish and Fisheries, 2014
https://www.scopus.com/pages/publications/84899073274?origin=scopusAI

50. DODGE-SAFE: Hybrid Dandelion Optimized HAF-BiLSTM for fishermen safety in the Gulf of Mannar Sesaiya, A., Marimuthu, J., Ramachandran, B., Thiraviam, S.G. Intelligent Decision Technologies, 2025
https://www.scopus.com/pages/publications/105013889283?origin=scopusAI

51. Automatic identification of maritime boundary alert system using GPS Kavitha, A., Ashok Kumar, N., Revathy, M. International Journal of Engineering and Technology(UAE), 2018
https://www.scopus.com/pages/publications/85082353038?origin=scopusAI

52. Comprehensive analysis of Li-Fi technology: positioning algorithms, security vulnerabilities, and future IoT applications Razaque, A., Bektemyssova, G., Khan, M., (...), Lloret, J. Telecommunication Systems, 2025
https://www.scopus.com/pages/publications/105019501103?origin=scopusAI

53. Edge-Intelligent Autonomous Buoy System for Decentralized Maritime Border Security Jenath, M., Samanta, S., Anand, U., Satwik, V. Proceedings of the 3rd International Conference on Innovations in Cybersecurity and Data Science, ICICDS 2026, 2026 https://www.scopus.com/pages/publications/105046076903?origin=scopusAI

54. Enhanced Fishermen Safety and Communication Using Machine Learning in Electromagnetic Water Networks Kalaivani, K., Jaya Adhithiya, D., Mukesh, G., (...), Nandhakumar, V. Proceedings - 3rd International Conference on Artificial Intelligence and Machine Learning Applications: Healthcare and Internet of Things, AIMLA 2025, 2025 https://www.scopus.com/pages/publications/105010560736?origin=scopusAI

55. Intelligent Maritime Boundary Protection System for Fishermen Utilizing GPS and Engine Control Balamani, T., Bharathi, R., Keerrthi, S.S., (...), Ranjana Priya, R. 2025 International Conference on Computing and Communication Technologies, ICCCT 2025, 2025 https://www.scopus.com/pages/publications/105009013014?origin=scopusAI

56. Fishermen helping system Devaki, P., Renuka, T., Sridhevi, S. International Journal of Recent Technology and Engineering, 2019 https://www.scopus.com/pages/publications/85060538250?origin=scopusAI

57. Protecting fishermen's by detecting and warning them while crossing sea borders using GSM and RFID technologies Kamalakannan, B., Naresh, K., Sakthivel, P. Proceedings of 2016 Online International Conference on Green Engineering and Technologies, IC-GET 2016, 2016
https://www.scopus.com/pages/publications/85020210077?origin=scopusAI

58. System for Offshore Communication and Tracking (OCTS) George, D.E., Dhanya, R., Cabral, D., (...), Krishnaraj, C.R. 2023 World Conference on Communication and Computing, WCONF 2023, 2023
https://www.scopus.com/pages/publications/85173066344?origin=scopusAI

59. Descriptive, correlation analysis and analytical hierarchy process of coastal community empowerment of Bengkulu City, Indonesia Cahyadinata, I., Nusril, Gushevinalti International Journal on Advanced Science, Engineering and Information Technology, 2020 https://www.scopus.com/pages/publications/85087945195?origin=scopusAI

60. Blue Ocean: A Digital Hub for Empowering India's Fishermen Vibin, S.R., Sherin, M.P., Shanker, S. 2025 2nd International Conference on Intelligent Technologies for Sustainable Electric and Communications Systems, iTech SECOM 2025, 2025 https://www.scopus.com/pages/publications/105032525653?origin=scopusAI

61. Blue crimes and ocean harmscapes: strategies for tackling Transnational Maritime Environmental Crimes in the Global South Hübschle, A., Lindley, J. Frontiers in Conservation Science, 2024 https://www.scopus.com/pages/publications/85207007902?origin=scopusAI

62. The Impact of Women's Empowerment Innovations through Integrated Education-Facilitative in Parepare City Abdullah, B., Yusuf, S. Pakistan Journal of Life and Social Sciences, 2024 https://www.scopus.com/pages/publications/85203332427?origin=scopusAI

63. Women Empowerment Status in the Coastal Fishing Communities of Bangladesh Rahman, M.W., Haque, A.B.M.M., Zaman, T., (...), Nazia, T. SAGE Open, 2024 https://www.scopus.com/pages/publications/85193701186?origin=scopusAI

64. Integrated Multi-Hazard Early Warning Systems: Technologies, Governance, and Implementation Challenges Fadiya, N., Sekti, B.A. Building Resilience through Multi-Hazard Early Warning Systems, 2026 https://www.scopus.com/pages/publications/105039735711?origin=scopusAI

65. Open and interoperable maritime surveillance framework set to improve sea-borders control Erbas, C., Tuncer Cetin, F., Yilmaz, B., (...), Bulca, A. Communications in Computer and Information Science, 2012 https://www.scopus.com/pages/publications/84867667204?origin=scopusAI

66. Towards the adoption of AI, IoT, and Blockchain technologies in Bangladesh's maritime industry: Challenges and insights Tahsin, R., Rantu, S.B.A., Rahman, M., (...), Karim, M.R. Results in Engineering, 2025 https://www.scopus.com/pages/publications/85213208839?origin=scopusAI

67. Examining ICT Innovation for Sustainable Terminal Operations in Developing Countries: A Case Study of the Port of Radès in Tunisia Sahraoui, A., Tran, N.K., Tliche, Y., (...), Taghipour, A. Sustainability (Switzerland), 2023 https://www.scopus.com/pages/publications/85163123578?origin=scopusAI

68. Border and Weather Alert Security System for Fishermen Ajith, A.S., Dhotre, O.S., Hegde, K.P., (...), Snehalatha, V.M. Proceedings - 2023 7th International Conference on Design Innovation for 3 Cs Compute Communicate Control, ICDI3C 2023, 2023 https://www.scopus.com/pages/publications/85199915823?origin=scopusAI

69. Safety System for Fishing Vessels in Maritime Shelly, D.M., George, J., Shibu, M., (...), James, R. 2025 11th International Conference on Smart Computing and Communications: Intelligent Technologies for a Sustainable World, ICSCC 2025, 2025 https://www.scopus.com/pages/publications/105029616479?origin=scopusAI

70. IT implementation in developing country municipality: A sociocognitive analysis Sanford, C., Bhattacherjee, A. Journal of Global Information Management, 2007 https://www.scopus.com/pages/publications/47349125875?origin=scopusAI

71. International Maritime Boundary Alert for Fishermen Mary, W.S., Muthukumar, S., Suganya Pragathi, V., (...), Viiavaraghavan, R. Proceedings of the 2nd International Conference on Electronics, Communication and Aerospace Technology, ICECA 2018, 2018 https://www.scopus.com/pages/publications/85060879760?origin=scopusAI

72. COMPARATIVE LAW ENFORCEMENT MODEL AT SEA: LESSON LEARNED FOR INDONESIA Dewi, Y.K., Afriansyah, A., Darmawan, A.R. Indonesian Journal of International Law, 2020 https://www.scopus.com/pages/publications/85111849269?origin=scopusAI

73. The main challenges and barriers to the successful “smart shipping” Alop, A. TransNav, 2019 https://www.scopus.com/pages/publications/85073373757?origin=scopusAI

74. Bridging the maritime domain awareness gap: The role of new technologies in promoting equitable surveillance capabilities to enact environmental obligations under the law of the sea Eddine, R.A., Guliyeva, S. ASCOMARE Yearbook on the Law of the Sea, 2023 https://www.scopus.com/pages/publications/105003544089?origin=scopusAI